R FOR DATA ANALYSIS · R BASICS · LESSON 2

02 Text, numbers and TRUE or FALSE

The three basic data types in R, and the difference between one and two equals signs.

BY THE END OF THIS SECTION YOU WILL BE ABLE TO

- Recognize character, numeric and logical values
- Put text inside quotation marks
- Compare two values with a double equals sign

The R code in this lesson

```
# Character: text goes in quotes
"Ahmad"

# Numeric
30

# Logical: TRUE or FALSE
50 == 50   # TRUE
50 == 25   # FALSE
# 25 = 50 gives an error: one = means assign, not compare
```

What you will hear (times count from the end of the intro)

Intro Every value in R has a type. In this lesson, you will meet the three most common ones: text, numbers, and TRUE or FALSE. You will also see a small mistake that almost everyone makes at the start, and how to fix it. Let's take it slowly.

0:00 Here are our three notes from before: character, numeric, and logical. Let's look at each one.

0:10 First, character. Under the note, I wrote a name inside quotation marks. The quotes tell R: this is text. Without quotes, R would think it is the name of something else.

0:55 Next, numeric. A number, like thirty for an age, needs no quotes. R knows it is a number, and you can do math with it.

1:35 Now the logical type. Logical values are only TRUE or FALSE. We get them when we ask R a question, like: is twenty five equal to fifty?

1:52 I typed twenty five, one equals sign, fifty. And R shows an error in red. Please don't worry when you see red. It is R's way of helping us.

2:08 The problem is the single equals sign. In R, one equals sign means assign a value. To compare two values, we need two equals signs.

2:25 So I write fifty, two equals signs, fifty. Run it, and R says TRUE. Fifty is equal to fifty.

2:40 Then fifty, two equals signs, twenty five. R says FALSE. They are not equal.

2:55 So remember: text in quotes, numbers without quotes, and two equals signs to compare. Very well done.